

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Narappagari Sravya(192472192) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. **Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.**

Aim

To obtain exactly **2 gallons** of water in the **4-gallon jug** using a 4-gallon jug and a 3-gallon jug.

Problem Statement

Given two jugs of capacities 4 gallons and 3 gallons without measurement markings, determine how to get exactly 2 gallons in the 4-gallon jug.

Algorithm

1. Fill the 3-gallon jug.
2. Pour it into the 4-gallon jug.
3. Fill the 3-gallon jug again.
4. Pour water into the 4-gallon jug until it is full.
5. One gallon remains in the 3-gallon jug.
6. Empty the 4-gallon jug.
7. Transfer the remaining 1 gallon from the 3-gallon jug into the 4-gallon jug.
8. Fill the 3-gallon jug again.
9. Pour it into the 4-gallon jug.

Now the 4-gallon jug contains exactly **2 gallons**.

State Space Representation

Step	4-Gallon Jug	3-Gallon Jug
Start	0	0
Fill 3-gallon	0	3
Pour to 4-gallon	3	0
Fill 3-gallon	3	3
Fill 4-gallon from 3	4	2
Empty 4-gallon	0	2
Pour remaining to 4	2	0

Result

Exactly **2 gallons** are obtained in the 4-gallon jug.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.

i. What are the percept's for this agent?

ii. Characterize the operating environment.

iii. What are the actions the agent can take?

iv. How can one evaluate the performance of the agent?

v. What sort of agent architecture do you think is most suitable for this agent? Why?

Aim

To study the Mars Rover as an intelligent agent.

i. Percepts

- Camera images
- Rock and soil samples
- Temperature
- Atmospheric pressure
- Terrain information
- Chemical sensor readings
- Position (GPS-like localization)

ii. Operating Environment

- Partially observable
- Dynamic
- Sequential
- Uncertain
- Continuous
- Real-world environment

iii. Actions

- Move forward/backward
- Turn left/right
- Collect soil samples
- Drill rocks
- Capture images
- Analyze samples
- Transmit data to Earth

- Avoid obstacles

iv. Performance Measure

- Accurate sample collection
- Safe navigation
- Minimum energy consumption
- Maximum scientific discoveries
- Successful communication with Earth

v. Suitable Agent Architecture

Utility-Based Agent

Reason:

- Chooses the best action based on multiple objectives.
- Handles uncertainty.
- Optimizes scientific value and battery usage.
- Suitable for autonomous exploration.

3. **Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.**

Aim

To formulate the problem components of the 8-Queens problem.

Problem Formulation

Initial State

An empty 8×8 chessboard.

State Space

Any arrangement of queens on the board.

Operators

- Place one queen in an empty row.
- Move a queen within a row.

Goal State

Place 8 queens such that no two queens attack each other.

Constraints

- One queen per row.
- One queen per column.
- No two queens share a diagonal.

Search Strategy

Backtracking Search

Algorithm

Place queen in first row

If position is safe

 Place queen

Else

 Try next column

Repeat for all rows

If all 8 queens are placed

 Success

Else

 Backtrack

Result

A valid arrangement of eight queens is obtained without conflicts.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

Type of Problem Solving Agent

Goal-Based Agent

Reason:

The customer's goal is to reach the destination by selecting the most suitable cab.

Pseudocode

BEGIN

Input pickup location

Input destination

Display available cab types

Mini

Micro

Sedan

Prime

Shared

User selects cab

Calculate fare

Display estimated arrival time

If user confirms booking

 Book cab

 Assign nearest driver

 Display driver details

Else

Cancel booking

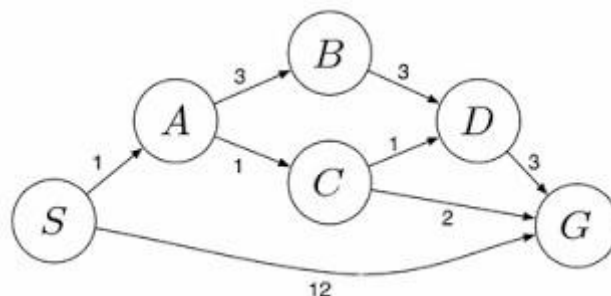
END

Result

The customer successfully books a suitable cab to reach the destination.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Aim

To determine the least-cost path from warehouse S to destination G using Uniform Cost Search.

Given Graph

Edges:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$

- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

Uniform Cost Search Algorithm

1. Insert start node into priority queue.
2. Expand the node having minimum path cost.
3. Add neighboring nodes with cumulative costs.
4. Repeat until goal node is selected.
5. Return the least-cost path.

Pseudocode

UCS(Graph, Start, Goal)

Create priority queue

Insert Start with cost 0

While queue is not empty

 Remove node with minimum cost

 If node is Goal

 Return path

 Expand neighbors

 Add neighbors with cumulative cost

End While

UCS Search Table

Expanded Node	Path	Cost
S	S	0
A	$S \rightarrow A$	1
C	$S \rightarrow A \rightarrow C$	2
D	$S \rightarrow A \rightarrow C \rightarrow D$	3
G	$S \rightarrow A \rightarrow C \rightarrow G$	4

Alternative paths:

- $S \rightarrow G = 12$
- $S \rightarrow A \rightarrow B \rightarrow D \rightarrow G = 10$
- $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G = 6$

Least-Cost Path

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost:

$$1 + 1 + 2 = 4$$

Result

The Uniform Cost Search algorithm finds the optimal path:

$S \rightarrow A \rightarrow C \rightarrow G$

Minimum Cost = 4